

# BENCHAO SU

School of Mathematical Sciences, Peking University  
[sbc@stu.pku.edu.cn](mailto:sbc@stu.pku.edu.cn)

## ACADEMIC INTERESTS

---

My research interest lies in the  $p$ -adic local Langlands program, especially in locally analytic representations coming from  $p$ -adic geometry.

## EDUCATION

---

### Peking University

- Ph.D. in Pure Mathematics (expected graduation: 2026).
- Advisor: Yiwen Ding.

## PUBLICATIONS AND PREPRINTS

---

4. **Locally analytic vectors in the completed cohomology of quaternionic Shimura curves** (joint with Zhenghui Li, Zhixiang Wu). arXiv:2601.13625.
3. **De Rham cohomology of Lubin-Tate spaces and Drinfeld spaces.** arXiv:2503.09352.
2. **On the locally analytic  $\mathrm{Ext}^1$ -conjecture in the  $\mathrm{GL}_2(L)$  case.** arXiv:2504.17683.
1. **Locally analytic vectors in the completed cohomology of unitary Shimura curves** (joint with Qiu Tian). arXiv:2505.10290.

## TALKS

---

8. *Locally analytic vectors in the completed cohomology of quaternionic Shimura curves*, Thematic Month, Around the Langlands program,  $p$ -adic aspects of the Langlands program, 2026.2, Marseille.
7. *On the Lie algebra cohomology of de Rham complex of Drinfeld tower*, Seminar at Claude Bernard Lyon 1 University and ENS de Lyon, 2025.11, Lyon.
6. *On admissibility of de Rham complex of Drinfeld tower of dimension 1 over  $L$* , Seminar at University Clermont Auvergne, 2025.11, Clermont-Ferrand.
5. *Solid locally analytic representations and solid almost quasi-coherent sheaves*, Recent progress in  $p$ -adic geometry, 2025.03, Yunnan.
4. *On Breuil's locally analytic  $\mathrm{Ext}^1$ -conjecture in the  $\mathrm{GL}_2(L)$  case*, ENTR Workshop 24, 2024.09, Essen.
3. *Translations and the locally analytic  $\mathrm{Ext}^1$  conjecture in the  $\mathrm{GL}_2(L)$  case*,  $p$ -adic geometry and analysis, 2024.06, Shenzhen.
2. *Translations and the locally analytic  $\mathrm{Ext}^1$  conjecture in the  $\mathrm{GL}_2(L)$  case*,  $p$ -adic Day, 2024.05, Beijing.
1. *Locally  $\sigma$ -analytic vectors in the completed cohomology of unitary Shimura curves*, Mini-conference on  $p$ -adic Langlands Program, 2023.08, Beijing.

---

<sup>1</sup>Updated March 24, 2026